

Lab 3: Component Modelling & Architectural Pattern Selection

Domain & SSL Certificate Expiry Alert System

Architecture Selection

We chose Layered Architecture for the Domain & SSL Certificate Expiry Alert System.

Architectural Style Analysis

Layered: presentation, business/application, integration and data layers with clear separation of concerns. Microservices: independently deployable services with independent scaling and fault isolation, but greater distributed-service and operational complexity. Client-Server: centralized control, but the monitoring workflow is concentrated in the central server.

Two Scenario-Related Reasons

1. Dashboard and access control are separated from asset management, scheduled monitoring, TLS/WHOIS scanning, alert escalation, notifications and auditing, directly mapping to FR-001 through FR-005.
2. TLS/WHOIS scanning is isolated behind a component interface so the Monitoring Engine can coordinate daily checks without exposing scanner implementation details. This supports modularity and high-volume monitoring.

Security Advantage

Role-based authorization can be enforced before administrative functions are used, while the Data Access Layer controls persistence instead of the presentation layer accessing the database directly. This supports NFR-002.

Performance Benefit

The Monitoring Engine and scanner isolate the scan workload from dashboard requests. This supports NFR-001: scan 1,000 domains and SSL endpoints in under 3 minutes.

Components

Monitoring Dashboard; Authentication & Authorization; Monitored Asset Manager; Monitoring Engine; Alert & Escalation Manager; Audit Trail Service; Notification Service; TLS/WHOIS Scanner; Alert Channels; External Targets; Data Access Layer; Monitoring Database.

Interfaces

Auth API; Asset API; Alert View; Audit View; TLS/WHOIS API; Alert API; Notify API; Audit API; DB Query.

Conclusion

The component model is traced to the actual Lab 1 Domain & SSL Certificate Expiry Alert System and exceeds the Lab 3 minimum of five components and four interfaces.